97

Good work!

Project:	Beneath Every Tide (BET)
Team No.:	2
Class:	CSE 3310; Fall 2026
Module:	Unified Modeling Language (UML)
Deliverable:	UML Document – Increment I delivery

Version: [3.0] **Date: [09/24/2026]**

Contributors:

Prasil Niraula
Jayneshkumar Patel
Hiya Kavalanekar
Alyssa Reyna Aguilar

Revision History

Version number	*Date*	*Originator*	*Reason for change*	*High-level description of changes*
1.0	09/11/2026	Team2	Initial draft	We all discussed and divided the parts for each team member. Then each team member drew their UML on paper.
2.0	9/18/2026	Team2	Review	Each team member reviewed each other's UML diagram and made necessary changes to it.
3.0	9/22/2026	Team 2	Final deliverable	Each team member converted their hand drawn UML to starUML and the facilitator compiled them together into this deliverable.

TABLE OF CONTENTS

1. INTRODUCTION AND PROJECT OVERVIEW 3

2. PROJECT CONTEXT DIAGRAM 4

3. UML DIAGRAMS 5

3.1 UML Diagram: "UML_Class_System" 5

3.2 UML Diagram: "UML_Activity_Setup" 6

3.3 UML Diagram: "UML_Activity_Win/Lose 7

3.4 UML Diagram: "UML_STD_Scoring" 8

3.5 UML Diagram: "UML_STD_AI" 9

3.6 UML Diagram: "Graphics" 10

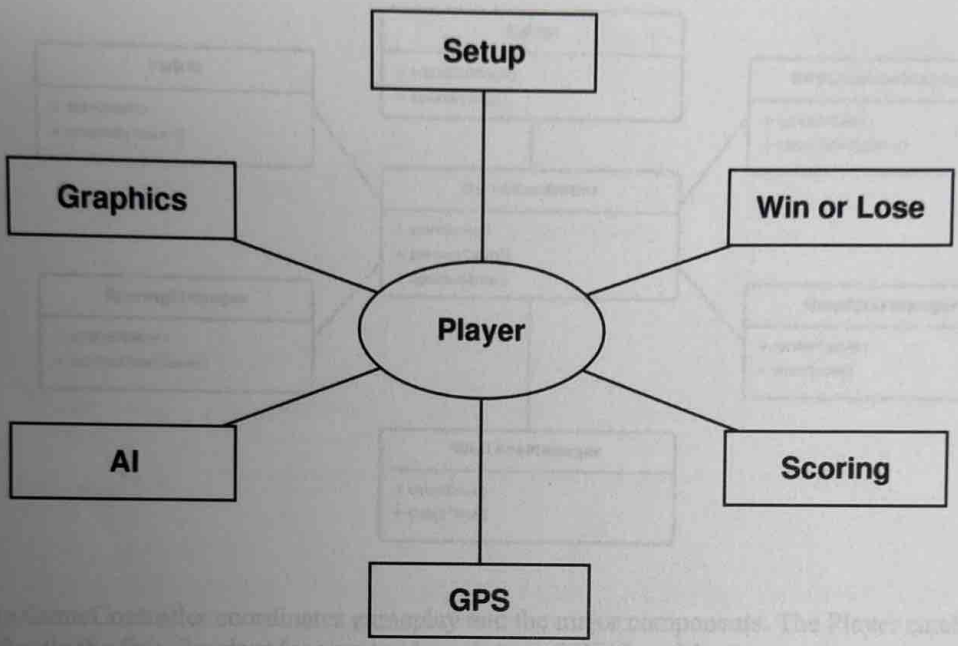
3.7 UML Diagram: "GPS Location" 11

1. Introduction and Project Overview

The game that will be discussed in this paper can be described as an Android fishing game where a user acts as a fisherman whose mission is to earn $50,000 to pay for his sister's cancer treatments. The game involves fishing and selling different fish in order to earn the money required. This narrative provides a personal reason for the fishing activity, thus adding significance to each fish caught.

There is a relatively simple artificial intelligence system in the game used to control the behavior of fish. The fishes in the game move, see the bait close to them, try to follow it, and finally bite or escape depending on the situation. This makes the game more interesting and requires some skills from users, such as timing and patience.

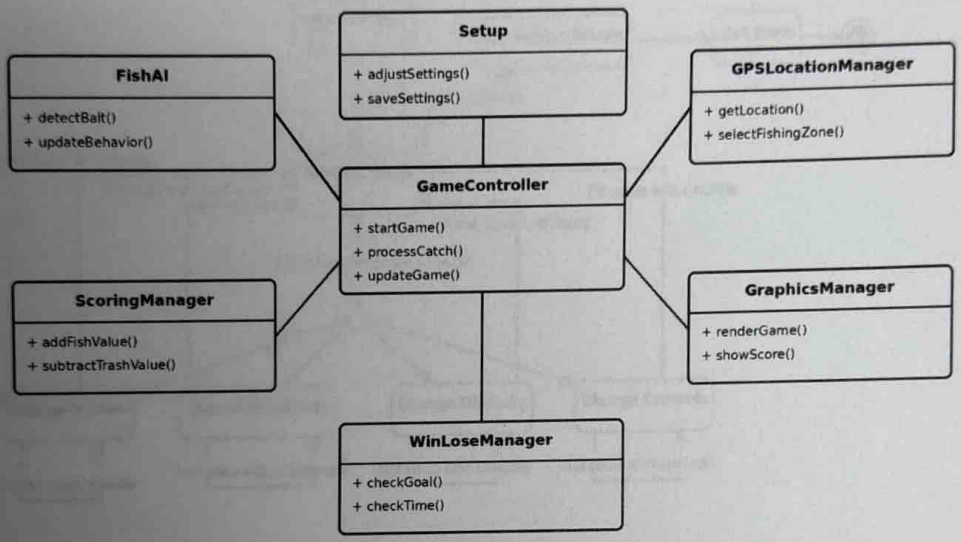
2. Project Context Diagram



The context diagram shows the player's interaction with the six major aspects of the game: the setup, graphics, AI, scoring, win/lose and GPS.

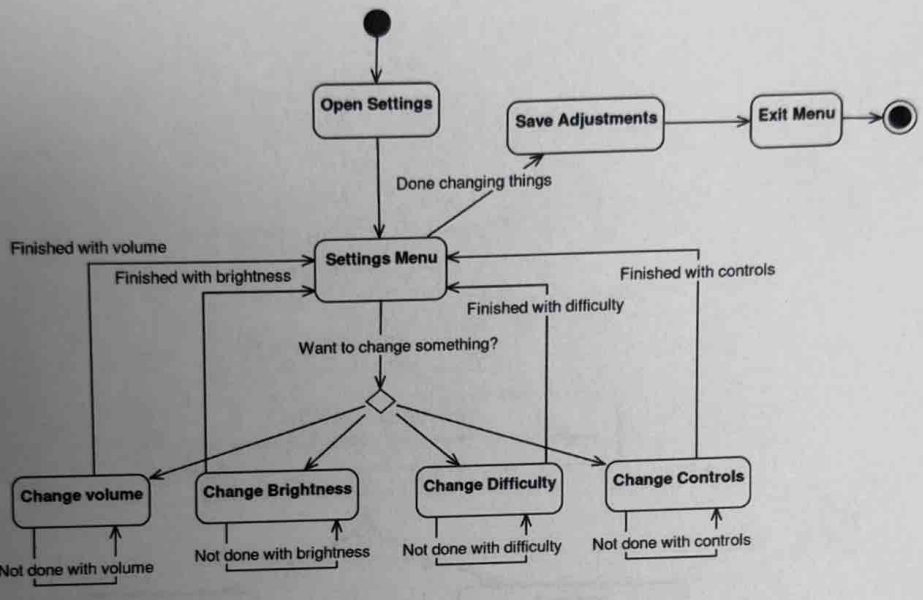
3. UML Diagrams

3.1 UML Diagram: "UML_Class_System"



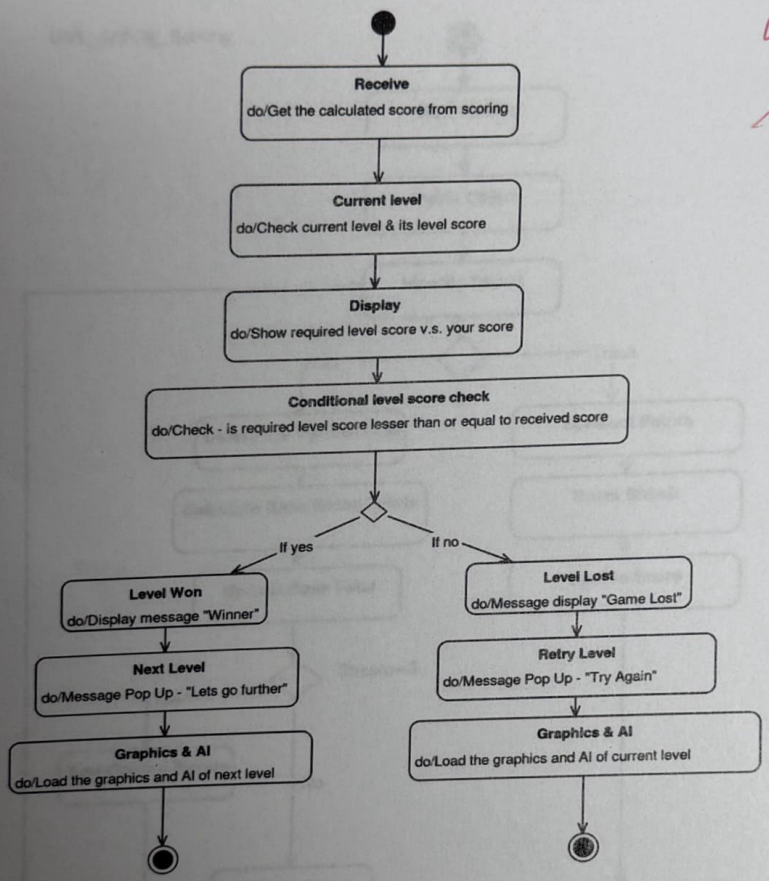
The GameController coordinates gameplay and the major components. The Player catches and sells the fish, ScoringManager tracks points, and WinLoseManager evaluates the result. GPSLocationManager can select a FishingZone, and GraphicsManager displays the game state.

3.2 UML DIAGRAM: "UML_ACTIVITY_SETUP"



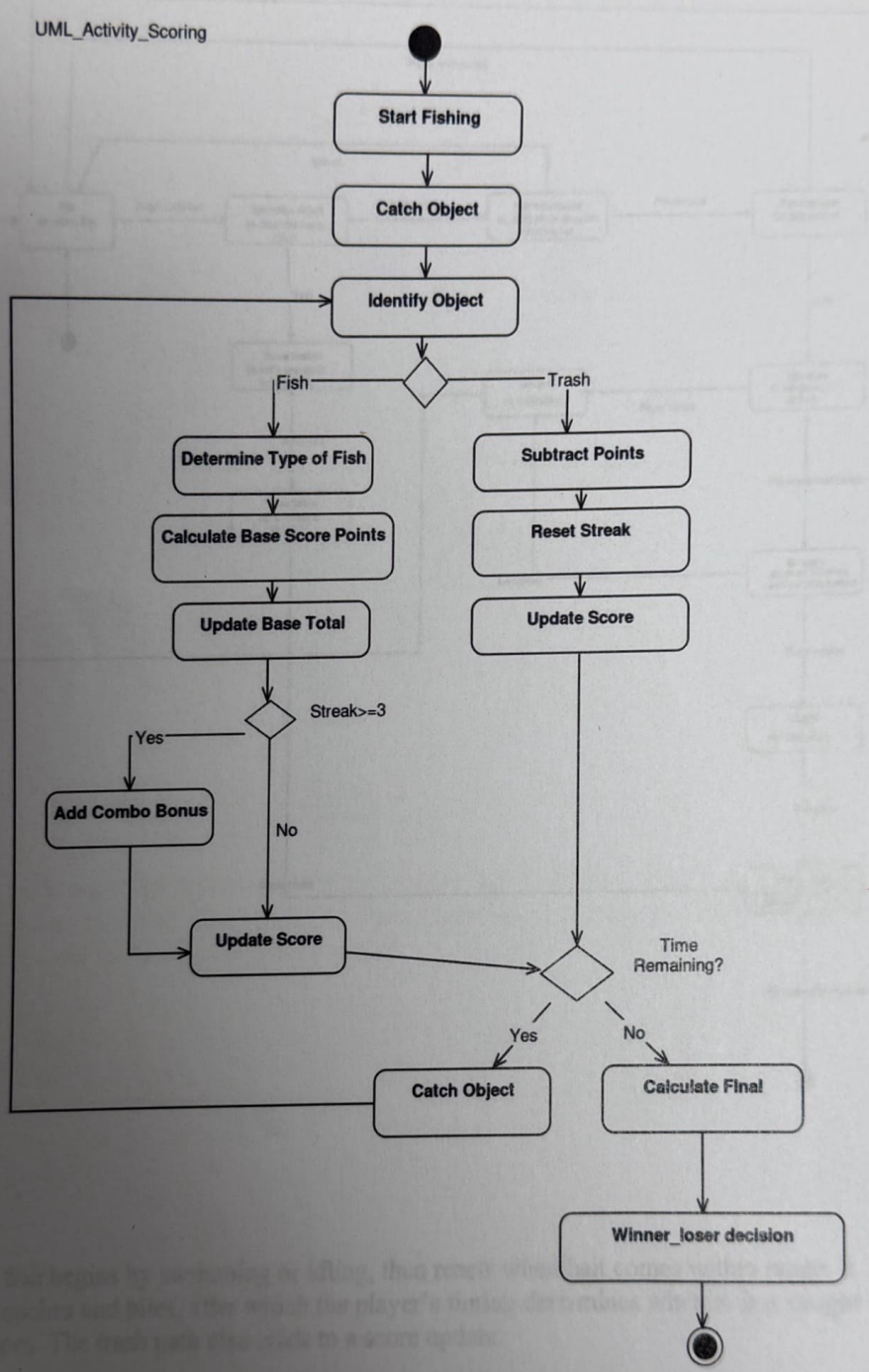
The setup flow lets the player change volume, brightness, difficulty, and controls. Each adjustment returns to the settings menu so the player can make another change before saving and exiting.

3.3 UML DIAGRAM: "UML_STD_WIN/LOSE



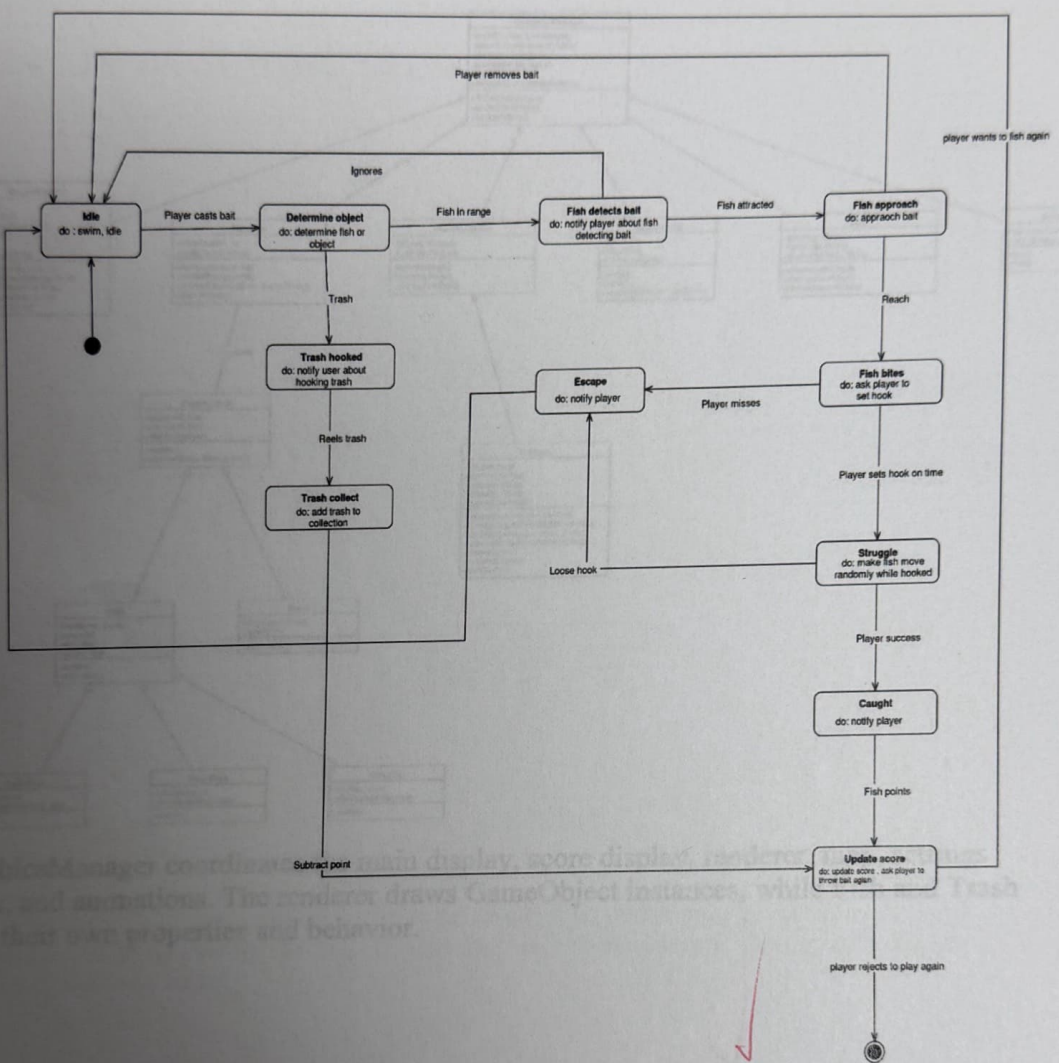
This diagram compares the received score with the requirement for the current level. Meeting the threshold advances to the next level; otherwise the player retries.

3.4 UML DIAGRAM: "UML_ACTIVITY_SCORING"



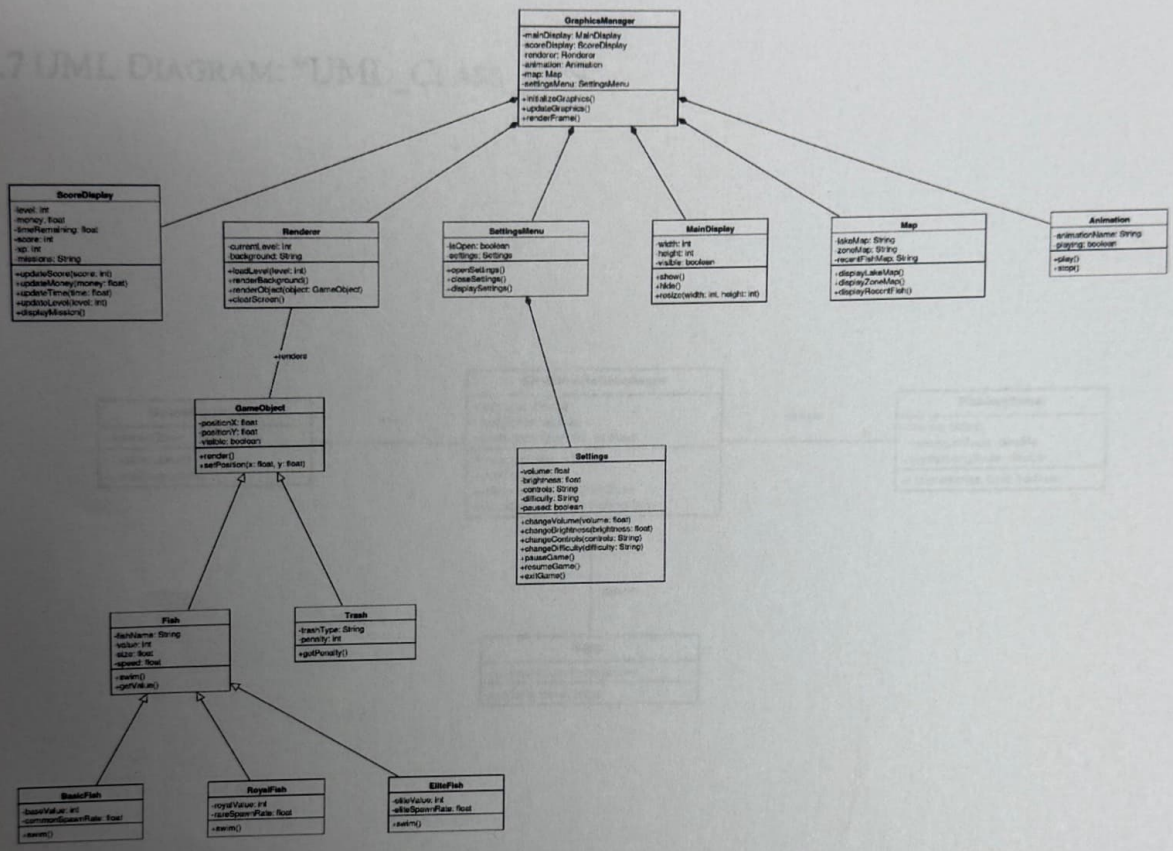
After an object is caught, the game distinguishes fish from trash. Fish adds base points and may earn a streak bonus; trash subtracts points and resets the streak. When time runs out, the final total is sent to the winner or loser decision.

3.5 UML DIAGRAM: "UML_STD_AI"



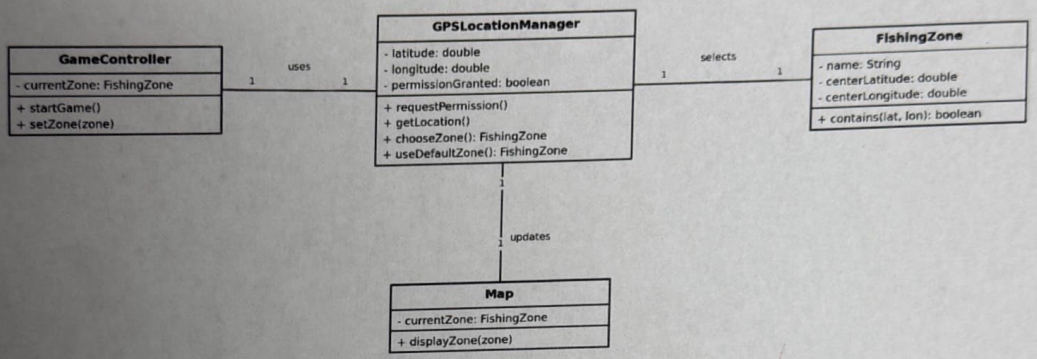
The fish begins by swimming or idling, then reacts when bait comes within range. It approaches and bites, after which the player's timing determines whether it is caught or escapes. The trash path also leads to a score update.

3.6 UML DIAGRAM: "UML_CLASS_GRAPHICS"



GraphicsManager coordinates the main display, score display, renderer, map, settings menu, and animations. The renderer draws GameObject instances, while Fish and Trash have their own properties and behavior.

3.7 UML Diagram: "UML_Class_GPS"



GPSLocationManager requests location permission, reads the player's coordinates, and selects a FishingZone for the map. If a location cannot be obtained, the default zone lets the player continue fishing.